# GaussianJW

## Parametrisation

The GaussianJW likelihood is a two-part likelihood for $\{(y, v)_i\}$. First a Gaussian observation $y$

$$y|\ldots \sim \mathcal{N}(p, V(p, n))$$

with mean (probability) $p$ and a variance function

$$V(p, n) = \exp\left(\beta_1 + \beta_2 \log(p(1-p)) + \beta_3 \log(n)\right)$$

and then an (conditional independent) observed variance $v$, where

$$\nu \frac{v}{V(p, n)}|\ldots \sim \chi^2_\nu.$$

The case $\beta_1 = 0, \beta_2 = 1, \beta_3 = -1$ resembles the case where a Binomial is approximated with a Normal. $(n, \nu)$ is considered as fixed.

## Link-function

The probability $p$ is linked to the linear predictor $\eta$ with a (default) logit link

$$p = \frac{1}{1 + \exp(-\eta)}$$

but other links are also possible.

## Hyperparameters

The hyperparameters are

$$\theta_1 = \beta_1$$
$$\theta_2 = \beta_2$$
$$\theta_3 = \beta_3$$

and the prior is defined on $(\theta_1, \theta_2, \theta_3)$.

## Specification

- `family="gaussianjw"`
- Required arguments: $y$, $n$ and $\nu$ (all vectors of the same length) as an `inla.mdata()`-object with this spesific ordering, see the example.

### Hyperparameter spesification and default values

**doc** `The GaussianJW likelihoood`

**hyper**

**theta1**

**hyperid** `65101`
**name** `beta1`
**short.name** `beta1`
**output.name** `beta1 for GaussianJW observations`

**output.name.intern** beta1 for GaussianJW observations
**initial** 0
**fixed** FALSE
**prior** normal
**param** 0 100
**to.theta** function(x) x
**from.theta** function(x) x

**theta2**

**hyperid** 65102
**name** beta2
**short.name** beta2
**output.name** beta2 for GaussianJW observations
**output.name.intern** beta2 for GaussianJW observations
**initial** 1
**fixed** FALSE
**prior** normal
**param** 1 100
**to.theta** function(x) x
**from.theta** function(x) x

**theta3**

**hyperid** 65103
**name** beta3
**short.name** beta3
**output.name** beta3 for GaussianJW observations
**output.name.intern** beta3 for GaussianJW observations
**initial** -1
**fixed** FALSE
**prior** normal
**param** -1 100
**to.theta** function(x) x
**from.theta** function(x) x

**survival** FALSE

**discrete** FALSE

**link** default logit probit

**pdf** gaussianjw

## Example

```
n <- 300
x <- rnorm(n, sd = 0.5)
eta <- 1 + x
p <- 1/(1 + exp(-eta))
df <- sample(10:100, n, replace = TRUE)
size <- df
```

```
va <- p * (1.0 - p) / size
v <- rchisq(n, df = df) * va / df
phat <- rnorm(n, mean = p, sd = sqrt(v))
Y <- inla.mdata(phat, v, size, df)

r <- inla( Y ~ 1 + x,
          data = list(Y = Y, x = x),
          family = "gaussianjw",
          ## this might be needed
          control.inla = list(cmin = 0))
summary(r)
```